

Healthcare Education: Program Completions

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AI Programming Foundations Project

Overview

This project builds a Python workflow to examine awards in nursing and related healthcare programs. It uses the NCES Integrated Postsecondary Education Data System (IPEDS), specifically the 2023 final/revised Completions file and institutional directory ([NCES completions data](#)). The notebook loads, cleans, groups, and visualizes the records without training a machine-learning model.

Dataset Description

The prepared CSV contains 37,828 records and 16 columns from CIP series 51, Health Professions and Related Programs, covering awards granted from July 1, 2022 through June 30, 2023. Each record represents an institution, program, major number, and award level rather than an individual student. Key fields include program code and name, award level, institution state, award counts, and reporting flags. Program and award names were linked from the official dictionary, while institution names and states were joined by UNITID from the directory. The extract covers 3,925 institutions before the analytical filters (National Center for Education Statistics [NCES], 2023).

Workflow Description

Pandas loads the local CSV and displays the first rows, missing-value counts, and record-key checks. The `normalize_labels` function trims text labels, and `select_comparable_records` selects first-major awards at detailed levels in the 50 states and Washington, DC. The `summarize_awards` function produces grouped totals and shares for programs, award levels, and institution states. Three Matplotlib bar charts and their accompanying interpretations communicate the results. NumPy checks that total awards match the sum of the reported men and women counts.

Key Decisions and Assumptions

The notebook combines code, documentation, results, and interpretation in one runnable document. This applies Danchev's (2022) approach to transparent, reproducible analysis. The CSV is kept beside the notebook so ordinary execution does not depend on a live download. Original counts and reporting flags remain unchanged. Wilson et al. (2017) recommend preserving source data and recording transformations in code; the cleaning functions return copies instead of overwriting the input dataframe.

Key Decisions and Assumptions (continued)

Whitespace normalization changes 1,091 labels. There are no missing values or duplicate record keys in the selected source fields, so no imputation or duplicate deletion is applied. The analytical selection removes 453 second-major records and 564 additional first-major records outside the geographical scope, leaving 36,811 rows from 3,838 institutions. The file already contains detailed award levels; the function explicitly allows those levels to guard against counting subtotals alongside their components. All 8,338 zero-award records in the analytical population remain. Zero means zero reported awards, not missing data or proof that a program ceased operating.

Results and Interpretation

The filtered records total 951,577 first-major healthcare awards. Figure 1 in the notebook shows the ten largest individual program categories. Registered Nursing/Registered Nurse leads with 250,576 awards, or 26.3% of the total. This is one program category; combining all nursing specialties would answer a different question.

Figure 2 compares qualification levels. Bachelor's degrees are the largest category, with 251,787 awards, or 26.5%. Certificates and degrees represent different training pathways, so one award does not imply the same training duration or workforce contribution at every level.

Figure 3 shows the ten states with the highest counts. California leads with 97,017 awards. The locations belong to institutions, not students or future employers. These raw counts are not population-adjusted and should not be interpreted as rankings of education quality or access.

Responsible Practice: Bias and Data Quality

The geographical filter excludes territories and freely associated states. That makes the scope consistent but means the findings cannot represent every location in the source. Restricting the analysis to CIP series 51 also excludes health-adjacent programs classified elsewhere. Dropping zero-award programs would raise average awards per record; treating missing counts as zero in a future release could lower totals. Those transformations are not performed here.

All total-award flags in this extract are R (reported), but reporting status does not eliminate possible reporting errors. Award counts do not identify unique graduates, licensure success, employment, or learning outcomes. A single reporting year cannot establish trends or causal effects. A next analysis could add population denominators, examine territories separately, and compare multiple years using consistent program definitions.

Reproducibility

The notebook ran top-to-bottom with 12 executed code cells, three embedded charts, and no saved cell errors. Grouped program, award-level, and state totals reconcile to 951,577 awards. The three chart images were visually reviewed. A macOS permission warning occurred during kernel shutdown after execution; it did not prevent the notebook from being saved.

The environment uses Python 3.12, with installed versions captured using pip freeze in requirements.txt. The README explains dependency installation and notebook execution. Danchev (2022) describes recording dependency versions to reduce differences caused by package updates. The local Git repository contains main and development branches, with separate commits for dataset preparation, notebook ingestion, and the cleaning/analysis stage, consistent with version-control guidance from Wilson et al. (2017). Publication to GitHub remains a separate step; no public repository is claimed here.

References

Danchev, V. (2022). Reproducible data science with Python: An open learning resource. *Journal of Open Source Education*, 5(56), 156. <https://doi.org/10.21105/jose.00156>

National Center for Education Statistics. (2023). *Integrated Postsecondary Education Data System: 2023 Completions (C2023_A_RV), institutional directory (HD2023), and Completions dictionary* [Data sets and data dictionary]. U.S. Department of Education. Completions final/revised release: September 2025. [Completions CSV archive](#); [institutional directory](#); [data dictionary](#).

Wilson, G., Bryan, J., Cranston, K., Kitze, J., Nederbragt, L., & Teal, T. K. (2017). Good enough practices in scientific computing. *PLOS Computational Biology*, 13(6), e1005510. <https://doi.org/10.1371/journal.pcbi.1005510>